

Machine Learning

A practical study reference with foundational concepts, examples, and common review questions.

1. Problem and Workflow

Machine learning fits a model from data to make predictions or discover structure. Features are inputs; a label or target is the value supervised learning predicts. Define the user decision and cost of mistakes before choosing a model.

Workflow: define a metric, inspect and clean data, split data, establish a baseline, train and tune on training/validation data, evaluate once on held-out test data, deploy, then monitor. Never let test data influence preprocessing or model selection: this is data leakage.

2. Learning Types and Models

Supervised learning uses labeled examples. Classification predicts categories; regression predicts numeric values.

Unsupervised learning finds structure without labels, such as clusters. Reinforcement learning learns actions from rewards and requires careful evaluation and safety constraints.

Models include linear/logistic regression, decision trees, random forests, gradient-boosted trees, support vector machines, clustering, and neural networks. Complexity is not automatically better; compare against a simple baseline.

3. Generalization and Metrics

Overfitting models training details that do not generalize; underfitting fails to capture useful patterns. Use representative data, regularization, validation, and simpler models when appropriate. Use time-ordered splits for time prediction and group-aware splits for correlated samples.

Accuracy can mislead on imbalanced data. Precision measures correctness among predicted positives; recall measures how many true positives were found; F1 balances them. MAE and RMSE evaluate regression errors. Choose metrics based on actual error costs.

4. Deployment and Common Questions

A deployed model needs versioned data and artifacts, monitoring, latency controls, security, and rollback plans. Track input drift and model quality as labels arrive. Review privacy, consent, bias, subgroup performance, and human oversight for high-impact decisions.

Classification predicts a category; regression predicts a number. Cross-validation evaluates multiple partitions. A baseline catches pipeline mistakes and shows whether added complexity provides useful improvement. Fit transformations only on training data, then apply the fitted pipeline to other splits.